

Worksheet 1

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Subject Name: Internet Of Things

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Subject Code: 25-CAH

1. Aim/Overview of the practical: (ODD UIDS)

Interface an Arduino Uno with LEDs to design the different patterns and follow the different cases. Use 6 different color LEDs and name them T1 to T6 and use resistors of suitable values.

- a) T1, T2 and T3 are in ON condition and T4, T5 and T6 are in Off conditions.
- b) T1, T2 and T3 are in OFF condition and T4, T5 and T6 are in ON conditions.
- c) T1, T3 and T5 are in ON conditions and T2, T4 and T6 are in OFF conditions.
- d) T1, T3 and T5 are in OFF conditions and T2, T4 and T6 are in ON conditions.

2. Apparatus:

Hardware Requirements: Arduino Uno R3

Breadboard

LED

Jumper Wires

Software requirements: TinkerCAD

MS Windows 10

Arduino IDE

3. Circuit Diagram(TinkerCad):

- a) T1, T2 and T3 are in ON condition and T4, T5 and T6 are in Off conditions

Coding:

```
int led1 = 13;  
int led2 = 12;  
int led3 = 11;  
int led4 = 10;  
int led5 = 9;  
int led6 = 8;
```

```
void setup(){  
  //setting all pins as output  
  pinMode(led1 , OUTPUT);
```

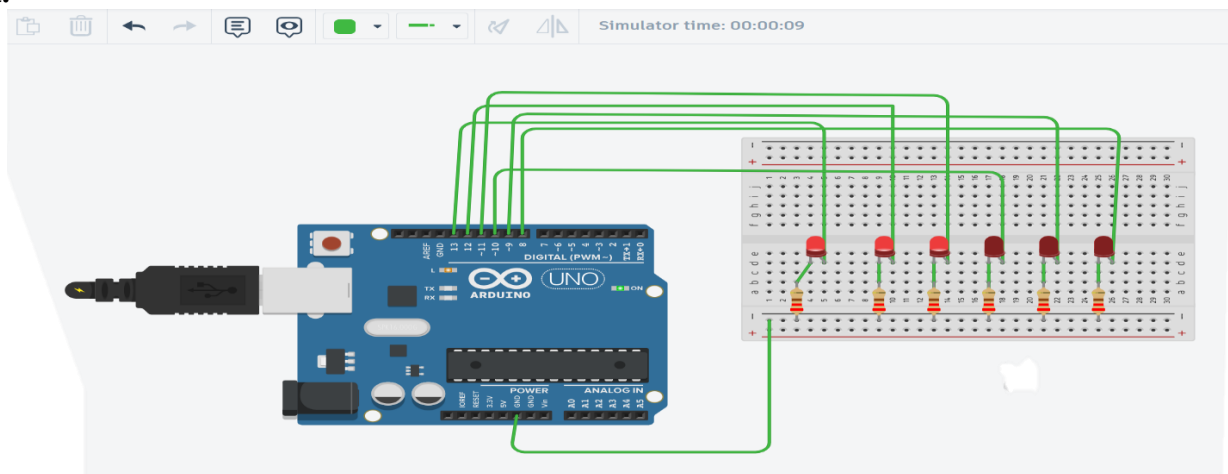
```
pinMode(led2 , OUTPUT);
pinMode(led3, OUTPUT);
pinMode(led4 , OUTPUT);
pinMode(led5 , OUTPUT);
pinMode(led6 , OUTPUT);
}
```

```
void loop(){
  //First led 3 ON
  digitalWrite(led1 , HIGH);
  digitalWrite(led2 , HIGH);
  digitalWrite(led3 , HIGH);

  //LAST 3 OFF
  digitalWrite(led4 , LOW);
  digitalWrite(led5 , LOW);
  digitalWrite(led6 , LOW);

  //DELAY
  delay(2000);
}
```

Output:



b) T1, T2 and T3 are in OFF condition and T4, T5 and T6 are in ON conditions.

Coding:

```
int led1 = 13;
int led2 = 12;
int led3 = 11;
int led4 = 10;
int led5 = 9;
int led6 = 8;
```

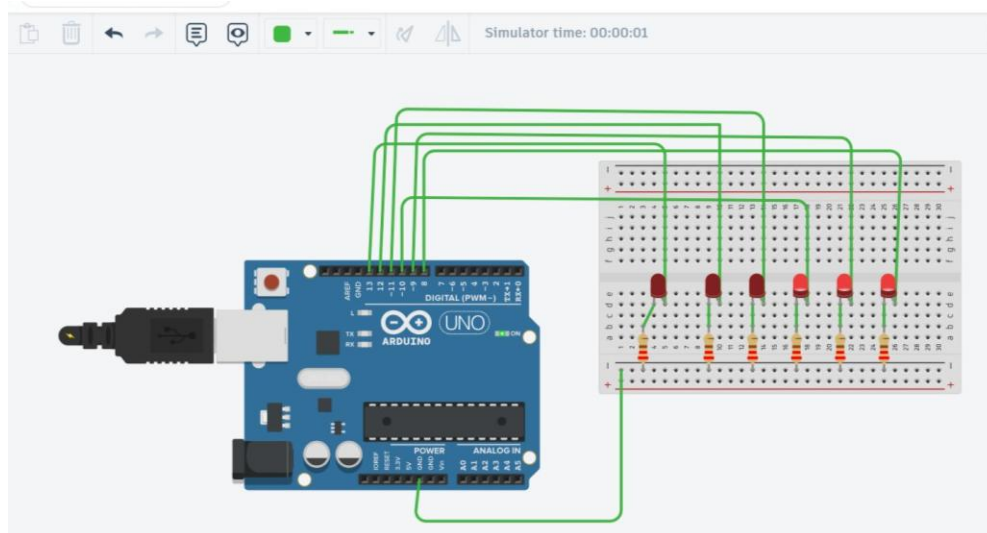
```
void setup(){
  //setting all pins as output
  pinMode(led1 , OUTPUT);
  pinMode(led2 , OUTPUT);
  pinMode(led3, OUTPUT);
  pinMode(led4 , OUTPUT);
  pinMode(led5 , OUTPUT);
  pinMode(led6 , OUTPUT);
}
```

```
void loop(){
//FIRST 3 OFF
  digitalWrite(led1, LOW);
  digitalWrite(led2 ,LOW);
  digitalWrite(led3 , LOW);

//LAST 3 ON
  digitalWrite(led4 , HIGH);
  digitalWrite(led5 , HIGH);
  digitalWrite(led6 ,HIGH);

  delay(2000);
}
```

Output:



- c) T1, T3 and T5 are in ON conditions and T2, T4 and T6 are in OFF conditions.

Coding:

```
int led1 = 13;
int led2 = 12;
int led3 = 11;
int led4 = 10;
```

```
int led5 = 9;
int led6 = 8;

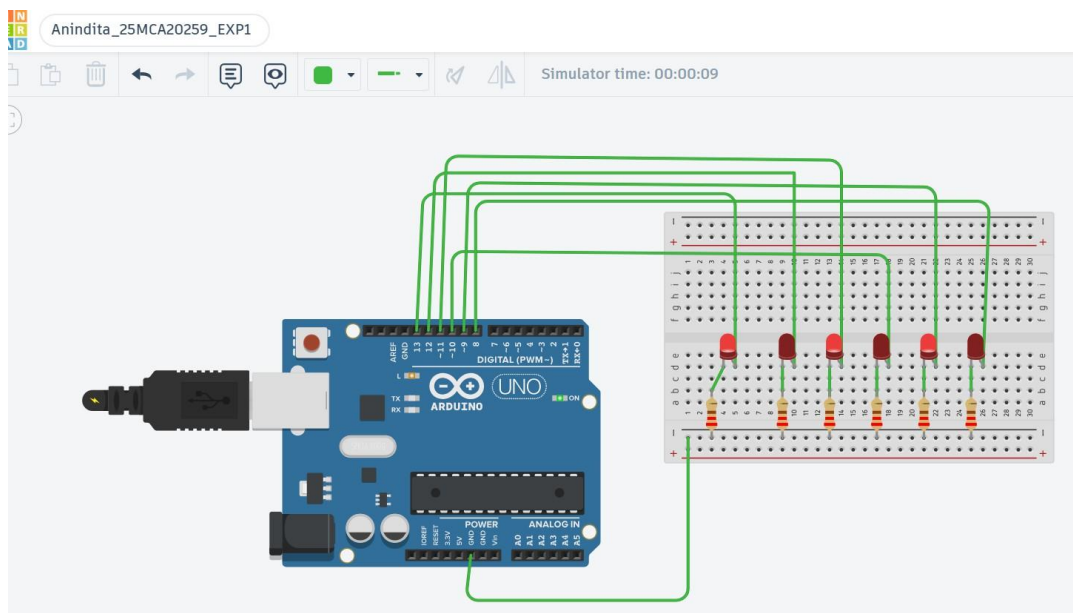
void setup(){
  //setting all pins as output
  pinMode(led1 , OUTPUT);
  pinMode(led2 , OUTPUT);
  pinMode(led3, OUTPUT);
  pinMode(led4, OUTPUT);
  pinMode(led5 , OUTPUT);
  pinMode(led6 , OUTPUT);
}
```

```
void loop(){

  digitalWrite(led1, HIGH);
  digitalWrite(led2 ,LOW);
  digitalWrite(led3 ,HIGH);

  digitalWrite(led4 , LOW);
  digitalWrite(led5 , HIGH);
  digitalWrite(led6 ,LOW);
```

Output:



d) T1, T3 and T5 are in OFF conditions and T2, T4 and T6 are in ON conditions.

Coding:

```
int led1 = 13;
```

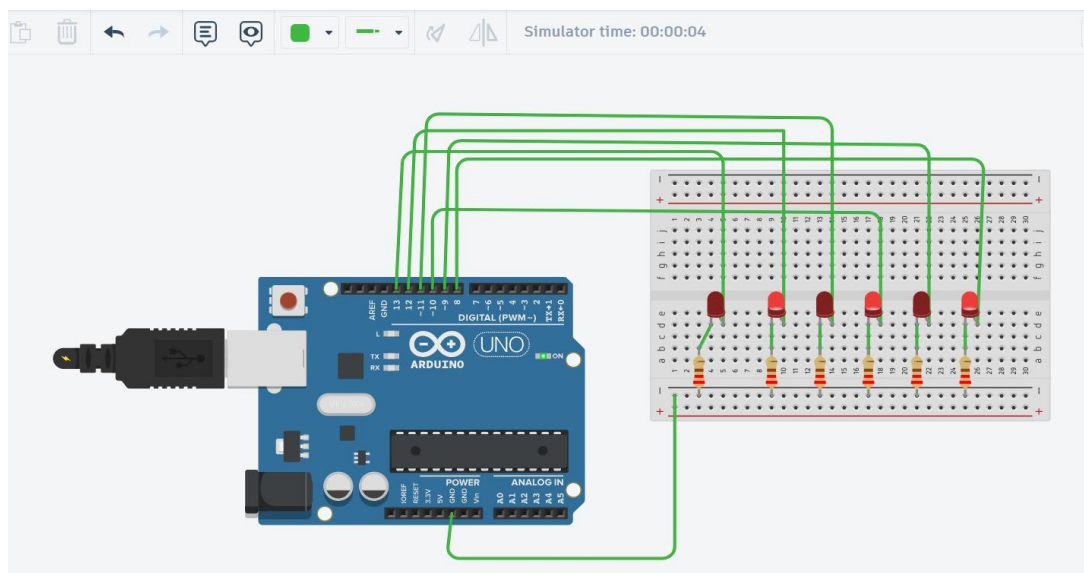
```
int led2 = 12;  
int led3 = 11;  
int led4 = 10;  
int led5 = 9;  
int led6 = 8;
```

```
void setup(){  
  //setting all pins as output  
  pinMode(led1 , OUTPUT);  
  pinMode(led2 , OUTPUT);  
  pinMode(led3 , OUTPUT);  
  pinMode(led4 , OUTPUT);  
  pinMode(led5 , OUTPUT);  
  pinMode(led6 , OUTPUT);  
}
```

```
void loop(){  
  
  digitalWrite(led1, LOW);  
  digitalWrite(led2 ,HIGH);  
  digitalWrite(led3 ,LOW);  
  
  digitalWrite(led4 , HIGH);  
  digitalWrite(led5 , LOW);  
  digitalWrite(led6 , HIGH);
```

```
  delay(2000);  
}
```

Output:



5. Learning outcomes :

- 1.Components of Arduino Uno R3.
- 2.Connecting breadboard , Arduino Uno R3, and led by using Jumper Wires.
- 3.Writing C++ code to customize working of LEDs in circuit using TinkerCAD.